PORTFOLIO · 2026-11

Iván Rodrigo Pianto Antón

Full stack software engineer · AI systems architect · Tech lead

I build software for real operations: multi-tenant ERPs, e-commerce platforms run by AI agents, civil-engineering tools and hydrological risk models.

Piura, Peru

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01

About

I combine a civil-engineering background with full stack architecture, and I work with discipline: specs before code, tests first, adversarial review and honest metrics.

KEY STACK

TypeScript, Python, Next.js 16, NestJS, FastAPI, PostgreSQL

LANGUAGES

Spanish — native

English — professional technical

EDUCATION

Civil Engineering and Information Technology — university education (Piura, Peru)

02

By the numbers

13,223
tests in the release gate of a multi-tenant ERP

20,491
ads analyzed with AI for USD 25.53

147k
stamps in a catalog rebuilt by scraping

25,290
river reaches in a national flood-hazard run

04

Featured work

Eight projects in full detail. Client and company names are anonymized.

01 ● PLATFORMS AND SAAS

Multi-tenant ERP and traceability for agro-export

Peruvian seafood exporter, through a software company · Tech lead and architect · 2024 – present

SaaS platform covering intake, production, warehouse, foreign trade and e-invoicing. A web panel, an offline mobile app and WhatsApp bots work on the same backend.

- Document engine with AI extraction, validation rules and a human-review traffic light; offline-first mobile app with an SQLite queue.
- Full foreign-trade area: proformas, containers, automatic packing proposals, a Kanban board and a historical load of 240 proformas (~USD 49.9M).
- Production with batch kardex, GS1 traceability and shrinkage reconciled against the plant's Excel (457 waste guides imported with zero rejections).
- SUNAT e-invoicing (invoices, receipts, credit/debit notes and dispatch guides) through a decoupled adapter.
- WhatsApp bots that log arrivals and issue proformas from text or voice notes, in production since August 2026.
- Release gate with 13,223 tests, Prometheus and Grafana observability, and infrastructure cut to ~USD 83/month.

STACK

NestJS · Fastify · PostgreSQL 17 · Drizzle · Redis · BullMQ · Next.js 16 · React 19 · Expo · n8n · Gemini · Whisper · Docker · GitHub Actions · Prometheus · Grafana

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02 ● AI AND AGENTS

E-commerce platform run by AI agents

Latin American e-commerce startup · Lead full stack developer and product architect · 2024 – present

Platform for direct-to-consumer, cash-on-delivery sellers: market research, offer strategy, creatives, landing pages, orders and AI-assisted sales.

- v1: an 11-agent AI pipeline (DAG) with 2 human checkpoints that validates a product and produces avatar, competitors, branding, copy, creatives and a landing page in ~30–45 min.
- v2: rebuilt as a multi-tenant SaaS in 20+ chained changes: identity, credit-based billing, financial engine, landing pages, publishing, orders and shipments.
- WhatsApp CRM with an AI sales agent that understands voice notes and photos, automatic follow-ups, courier integration and AI spend control.
- Automated creative studio: 17 day-one ads with audio QA (−14 LUFS) and render time cut from 346 to 199 s per ad.
- Suites of up to 2,120 tests plus Playwright E2E, and a blind dual review before every merge.

STACK

Next.js 16 · React 19 · TypeScript · PostgreSQL · Cloudflare Workers · D1 · R2 · OpenRouter · Anthropic API · Whisper · Remotion · Playwright · Vitest · Mercado Pago

03 ● AI AND AGENTS

Niche radar: AI ad-market intelligence

Module of the e-commerce platform · Architect and developer · 2026

Collects public ads by niche and country, extracts signals with OCR, CLIP and Whisper, labels them with LLMs and ranks winning ads with a survival score.

- Parallel collector with 20–30 workers: 923–1,717 ads per minute versus 68 sequentially.
- 20,491 ads analyzed for USD 25.53 and 62,581 phrases clustered for USD 0.40.
- Kaplan–Meier survival ranking and transcription on spot VMs (10 videos for USD 0.0048).
- Hexagonal architecture, multi-tenant isolation and deployment with backup and a rollback script.

STACK

Python 3.13 · FastAPI · Huey · DuckDB · Parquet · CLIP · Whisper · OCR · OpenRouter · GCP spot VMs · ECharts

04 ● ENGINEERING AND DATA SCIENCE

Construction company platform: procurement, estimating and tenders

Construction contractor in northern Peru · Engineer and developer · 2026

Internal platform and engines covering procurement, cost estimating, tender packages and works control, plus an AI assistant for the engineers.

- Internal platform with suppliers validated against a registry of 18.5M tax IDs, requests for quotation, price comparison and 13 corporate templates.
- Purchase-invoice reading: UBL XML, QR, text and LLM extraction, with automatic assignment to works (63 cases covered).
- Unit-price and quantity-takeoff engine with a hierarchical predictor: 13.8% median error on blind sets, over a corpus of 888 public works.
- Works portal with earned value (CPI, EAC) and a quality dossier; 10+ quotes in two weeks, up to S/ 237k, read from DWG/DXF drawings.
- WhatsApp assistant for the engineers, with rules outside the model, voice notes and sandboxed jobs.

STACK

Python · FastAPI · SQLAlchemy · PostgreSQL · Alpine.js · LibreOffice headless · ezdxf · Playwright · faster-whisper · systemd · Cloudflare Tunnel

05 ● ENGINEERING AND DATA SCIENCE

Road intelligence platform for Peru

Own project · Engineer and developer · 2026

Traffic, pavements, urban corridors and road-work costs on open data, with a modular per-country design.

- Market sizing from open public-procurement data: road consultancy is worth ≈ S/ 450M a year.
- National traffic model (143k km network, 253 zones, ODME against 73 toll stations): typical travel-time error from 55.6% to 13.0%.
- Urban corridors for 12 cities calibrated to a 6.6% out-of-sample MAPE by day type.
- Cost and solutions engine backtested on real project files with pre-registered rules; norms engine with 355 tests.
- 3D web viewer and automatically generated DOCX technical documents.

STACK

Python · FastAPI · OSMnx · AequilibraE · scikit-learn · MapLibre · Chart.js · python-docx · pytest

06 ● ENGINEERING AND DATA SCIENCE

Flood and debris-flow early warning

Own project, piloted in Piura · Engineer and data scientist · 2026

Impact-based early warning system for Peru, from forecast rainfall to thresholds at bridges and river reaches.

- Ensemble rainfall → LSTM discharge → Gumbel thresholds at 6 bridges: NSE 0.793 versus 0.597 for the baseline, and a Brier skill score of 0.60.
- National run over 25,290 river reaches with an automatic quality gate.
- Honest validation: the classic 20 mm rule had no skill (POD 0.03–0.16); the new debris-flow model reached TSS 0.81 on a locked 2025–26 set.
- Regional LSTM over 136 basins with routing (KGE from 0.17 to 0.52) and a public web viewer.

STACK

Python · PyTorch · ECMWF ENS · GloFAS · PISCO · CHIRPS · HydroBASINS · FastAPI

07 ● AI AND AGENTS

Live meeting copilot with on-device speech recognition

Own product · Author and architect · 2026

Listens to a meeting live, transcribes on the CPU and flags open questions and conflicts with earlier decisions. Productized with accounts, subscriptions and an MCP client.

- Streaming ASR on CPU with a 0.18–0.21 real-time factor, validated on a real 2 h 23 min meeting.
- Found and fixed a use-after-free in the voice detector that corrupted 5.5% of timestamps.
- Three lenses: semantic matching, suggested questions and conflicts against a ledger of 22 decisions; 7-model benchmark.
- Google login, API tokens stored only as hashes, subscriptions and a public MCP client; from 150 to 652 tests with zero warnings.

STACK

Python · FastAPI · ONNX Runtime · sherpa-onnx · Silero VAD · e5 embeddings · SQLite · MCP · Playwright

08 ● AI AND AGENTS

Mobile cockpit for AI coding agents (open source)

Own open-source project · Author and architect · 2025 – present

Mobile terminal and chat to drive coding agents running on your own computer: transcripts, tool approvals, notifications and an encrypted relay.

- Monorepo with a zod-validated shared protocol, a WebSocket and tmux daemon, a PWA and a native Expo app.
- QR pairing with end-to-end encryption (Noise IKpsk2) and a containerized relay with self-hosted push.
- Git, PR and CI panel, host diagnostics, themes and an adaptive tablet and desktop layout.
- 2,600+ tests across packages (server 889, mobile 1,530, protocol 231).

STACK

TypeScript · React · Expo · React Native · Vite · xterm.js · node-pty · WebSocket · Noise protocol · Docker · nginx · Playwright

05

All projects

17 projects across 4 areas, numbered as in the featured work.

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Module of the e-commerce platform · 2026

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Own project · 2026

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Own product · 2026

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Own open-source project · 2025 – present

09 ● PLATFORMS AND SAAS

Stamp collector platform and marketplace

Own project · 2026

Catalog of ~147k stamps, real-time auctions, collector accounts and AI enrichment, all on the edge.

- Catalog grown from 10k to ~147.5k stamps with a two-phase scraper: persistence from 0% to 95.4% and 97.6% image coverage.
- Auctions with atomic bids and optimistic concurrency; vector search and identification.
- Security hardening: removed a public SQL endpoint and fixed SSRF, bid impersonation and OAuth account pre-hijacking.

STACK

Next.js 16 · Cloudflare Workers · D1 · R2 · Vectorize · Workers AI · Playwright · Vitest · n8n

10 ● PLATFORMS AND SAAS

Personal finance engine for AI agents

Own project · 2025 – present

Command-line ledger that an AI agent can operate: imports bank emails and statements, deduplicates, classifies and budgets.

- Parsers for bank emails and PDF/XLSX statements with two-way deduplication.
- Weekly envelope budgeting, a softmax classifier with guards, net worth and FX.
- Hardened local dashboard (Origin, Sec-Fetch-Site, CSP and single writer); 212 tests.

STACK

Python · SQLite · openpyxl · Gmail API · pytest · uv

11 ● AI AND AGENTS

Personal AI assistant and automations

Personal use · 2026

Telegram assistant running as a service, with custom skills, scheduled monitors and control of coding-agent sessions.

- Skill that issues a recurring tax receipt through browser automation, with a double guard (confirmation + single-use token) added after a real incident.
- Agent-session supervisor through hooks and tmux; the first version shipped with 182 tests.
- Model routing by cost and latency with a fallback chain, persistent memory and a cap of 3 proactive messages a day.

STACK

Python · Telegram Bot API · systemd · cron · tmux · Chrome DevTools Protocol · OpenRouter · SQLite

12 ● AI AND AGENTS

MCP server for AI image and video generation

Local fork of an open-source project · 2026

MCP server that drives a web-based AI generation app through the Chrome DevTools Protocol, shared by several agents.

- Video prompt compiler and end-to-end generation verified with a pre-approved paid test.
- Full repair after a UI change: from 5 working and 8 broken functions to 18 and 0; 235 tests.
- Kept out of upstream because a selector map contained personal data.

STACK

Node.js · Playwright · Chrome DevTools Protocol · MCP · systemd · Xvfb

13 ● WEB, BUSINESS AND MORE

Vehicular-combat racing game for Roblox

Own project, inspired by a cult film · 2026

Game built code-first on Linux, with systems design, an economy and competitive multiplayer.

- Design document with 36 logged decisions, plus a balance and economy sheet.
- 12 phases: progression, arsenal, story, ranked play, monetization, live ops, localization and accessibility.
- Server-authoritative design and 921 unit tests with clean lint, formatting and types.

STACK

Luau · Rojo · Lune · selene · StyLua · luau-lsp · Roblox Studio MCP

14 ● WEB, BUSINESS AND MORE

Web, BI and process analysis for an exporter

Peruvian frozen-seafood exporter · 2026

B2B corporate website, executive dashboard and a survey of plant operations to specify the production module.

- Data-driven B2B site: interactive SVG cutting diagram, animated export-route globe and a filterable catalog.
- 12-tab executive dashboard: FOB sales, markets, prices, inventory, profitability, receivables and foreign-trade documents.
- Reverse-engineered the Excel-run operation (~306k formulas, Power Query and 991 lines of VBA) and mapped processes from 10 transcribed recordings (~60k words).

STACK

Next.js · Tailwind · Framer Motion · Python · Streamlit · pandas · Plotly · Power Query · VBA · Whisper

15 ● WEB, BUSINESS AND MORE

Bilingual company website with SEO and AI-generated media

Peruvian software and AI company · 2026

Brand site and client landing page positioned for search engines and AI answer engines.

- Client landing page rebuilt from mockups, with AI-generated WebP images and a vectorized logo.
- Brand SEO with JSON-LD, an ES/EN blog (3 + 3 articles) and a 26 s promo video with VideoObject markup.

STACK

HTML · CSS · JavaScript · JSON-LD · ffmpeg · Cloudflare Workers

16 ● WEB, BUSINESS AND MORE

Ad-tracker automation

E-commerce client in Peru · 2026

Fills the client's ad tracker from the media buyer's daily sheets.

- Join on a composite key (video code + delivery day) and handling of rows with several codes.
- Pure, testable core with thin I/O: from 46 to 99 tests, and a demo whose exact results were predicted and then verified live.

STACK

Google Apps Script · Google Sheets · Node.js tests

17 ● WEB, BUSINESS AND MORE

Operating-finance case: a USD 100M investment decision

University course · 2026

Financial model and two opposing theses on a sportswear multinational, built from its annual and quarterly filings.

- Formula-driven Excel model: FY19–FY26 history, FY27 projection, working-capital trap and the gap between free cash flow and dividends.
- Two complete theses (hold and buy), each with a 14-slide deck with native charts, a speaking script and 26–27 defense questions.
- JSON → PPTX/DOCX renderer with a content validator; it caught and corrected an inaccurate claim in the case itself.

STACK

Python · openpyxl · python-pptx · python-docx · Excel

06

How I work

01

Specs before code

Every large feature starts with a spec document that quotes the request verbatim, traceable tasks and a decision log.

02

Tests first

Strict TDD: red, green, refactor. Suites from hundreds to more than 13 thousand tests, with gates before every deploy.

03

Adversarial review

Blind dual review and lens-based review (risk, resilience, readability, reliability) on small, chained pull requests.

04

Honest metrics

Locked test sets, backtests with pre-registered rules and explicit reporting of what does not work yet.

05

AI with controlled cost

Model routing by cost, latency and quality, spend ledgers, and paid runs only with approval.

06

Security by design

Guards for irreversible actions, end-to-end encryption, agent sandboxing and personal-data hygiene.

07

Skills

FRONTEND AND MOBILE	TypeScript · React 19 · Next.js 16 · Tailwind CSS 4 · Expo · React Native · Vite · Alpine.js · MapLibre
BACKEND AND APIS	Node.js · NestJS · Fastify · Express · Python · FastAPI · WebSockets · MCP · BullMQ · Google Apps Script
DATA	PostgreSQL · SQLite · Cloudflare D1 · Redis · Drizzle · SQLAlchemy · DuckDB · pandas · Excel · Power Query · VBA · Streamlit
AI AND MACHINE LEARNING	Multi-agent orchestration · LLM document extraction · Speech to text (Whisper, ONNX) · Embeddings and vector search · PyTorch · scikit-learn · Model evaluation
CLOUD AND DEVOPS	Docker · nginx · GitHub Actions · Cloudflare · Vercel · Google Cloud · systemd · Prometheus · Grafana · Linux
QUALITY	TDD · Vitest · Jest · pytest · Playwright · Adversarial review
ENGINEERING AND DOMAIN	Cost estimating (unit prices, takeoffs) · Public procurement · Works control (earned value) · Traffic modelling · Hydrology and early warning · Foreign trade · E-invoicing · Financial analysis · Game design

08

Experience

2024 – present

Tech lead and systems & AI architect

Peruvian software and AI company

Architecture and development of a multi-tenant traceability and foreign-trade ERP in production, automated with AI agents.

2024 – 2025

Lead full stack developer

Latin American e-commerce startup

Frontend and backend architecture of an omnichannel CRM and ERP with WhatsApp messaging and payment gateways.

2025 – present

Open-source author and applied engineering

Independent

Tools for AI agents, civil-engineering software, and traffic and hydrology models.

09

Also

+ Webhook that answers Instagram and TikTok comments with HMAC signature verification.
+ Field photo inventory with OCR and vision, turned automatically into PDF and DOCX reports.
+ Traced Linux freezes down to a kernel bug and faulty RAM confirmed with Memtest86+.
+ ARM Android apps on an x86 tablet with Waydroid, cloud game servers on Docker, and a LaTeX CV.

10 · CONTACT

Let’s build something together.

Do you run an operation that needs software? Email me and tell me what you want to solve. Piura, Peru.

rodrigopianto2005@gmail.com github.com/R0DR0X CV (PDF) Portfolio (PDF)